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Improving digital skills for older adults: a game-based approach

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Improving Digital Skills for Older Adults: A Game-Based Approach

Abstract
The digital divide remains a persistent challenge for older adults, who often face structural, cognitive, and motivational barriers to effective participation in digital environments. This study presents and empirically evaluates a blended game-based assessment (GBA) and game-based learning (GBL) approach designed to support digital competences development in later life. Two complementary analogue-hybrid games, INFINITY and DiGiUP, were developed by operationalising the European DigComp 2.2 framework within concrete game mechanics. Motivational design was grounded in Self-Determination Theory (SDT). INFINITY focuses on baseline competence elicitation and confidence building through structured board-game interaction, while DiGiUP integrates scenario-based digital challenges to combine assessment and learning within gameplay.
The study was conducted within the Erasmus+ DIGIBLEND project and involved 45 older adults participating in facilitated small-group sessions in Slovenia. Digital competence outcomes were analysed using DigComp proficiency levels, and motivational experiences were measured using an SDT-based Gameplay Questionnaire. Quantitative analyses revealed statistically significant increases in demonstrated proficiency levels under DiGiUP compared to INFINITY across Information and Data Literacy, Safety, and Problem Solving. Qualitative observations and questionnaire responses further indicated high perceived competence, relatedness, and autonomy, suggesting strong motivational engagement in the game-based environment. While the exploratory design and single-country convenience sample limit generalisability and causal inference, the study provides design-informed empirical evidence that a blended GBA–GBL pathway can be feasibly implemented in real-world adult education settings. By translating competence frameworks into playable mechanics and embedding motivational principles within accessible formats, this work offers a replicable model for inclusive digital competence development among older adults.

Keywords: *digital competencies, game-based assessment, game-based learning, blended approach, older adults*

Abbreviations table

Abbreviation	Full term
GBA	Game-Based Assessment
GBL	Game-Based Learning
DESI	Digital Economy and Society Index
MDA	Mechanics-Dynamics-Aesthetics
SDT	Self-Determination Theory
GPQ	Game-Play Questionnaire
ICT	Information and Communication Technologies
DigComp	Digital Competence Framework for Citizens
EU	European Union
CTF	Capture the Flag
GDPR	General Data Protection Regulation
CFA	Confirmatory Factor Analysis

1. Introduction

Digital technology has become an integral part of modern life, facilitating communication, information access, and task management. With the widespread adoption of digital devices and the internet, digital competencies have become essential for almost everyone. These competencies do not only reflect the respective knowledge about information and communication technologies (ICT), but also the ability to use the technologies effectively and responsibly. Therefore, the development of digital competencies has been on the top of the European digital agenda (European Commission, 2024; Sala et al., 2020). *Digital competence* refers to the confident and critical use of ICT for work, leisure, and communication. According to the DigComp Framework (Vuorikari et al., 2022), digital competence encompasses five key areas: *Information and data literacy* **(**the ability to identify, locate, retrieve, evaluate, and use information effectively); *Communication and collaboration* (engaging in digital environments for collaboration, interaction, and participation); *Digital content creation* (creating, editing, and managing digital content); *Safety* (ensuring online privacy, security, and well-being); and *Problem solving* (innovatively resolving issues using digital tools).

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With more than a fifth (21.53%) of the EU population aged 65 and over in 2023, a figure projected to reach nearly 30% by 2050 (European Union, 2023;Corselli-Nordblad & Strandell, 2020) addressing the digital literacy of older adults is a critical societal priority. This demographic shift has highlighted a "second digital divide" rooted in technological skills and access rather than purely economic disparities (Reissmann et al., 2022). Currently, only 28% of European older adults possess basic digital skills, leaving a significant majority effectively excluded from digital culture (DESI, 2024). While the need for tailored learning opportunities is evident, efforts are often hindered by a notable reluctance among older adults to engage in traditional training, a challenge significantly more pronounced than in younger populations (Pedaste et al., 2023). Overcoming this barrier suggests the utility of innovative and inclusive approaches that not only make learning accessible but also foster motivation and confidence necessary for sustained digital adoption.

Traditional instructional methods often fall short in addressing the specific challenges of older learners, who benefit most from engaging, non-intimidating, and adaptable environments (Casselden, 2023). In this context, game-based assessment (GBA), the process of measuring competence through evidence gathered during play (Udeozor et al., 2024) and game-based learning (GBL), which leverages mechanics like feedback and progression to enhance instructional outcomes (Dubbels, 2013) emerge as promising solutions. By blending structured learning objectives with enjoyable game mechanics, these approaches create a motivating environment that encourages exploration and reduces the fear of failure (Kim & Ifenthaler, 2019). However, the literature reveals critical gaps: studies rarely focus on older adults in real-world service settings (Gutiérrez-Pérez et al., 2023), frequently address GBL and GBA in isolation, and often rely on fully digital delivery that may overwhelm those with low proficiency (Perim et al., 2024). We therefore adopt a blended GBA+GBL approach designed specifically for older adults. By using GBA to elicit evidence for profiling and GBL to embed guided practice through social interaction, this integrated pathway allows assessment to function as a confidence-building activity rather than a high-stakes test. The result is an incremental learning environment with facilitator support in a socially safe environment.

This study is situated within the European Commission's DigComp 2.2 framework (Vuorikari et al., 2022), specifically focusing on information and data literacy, safety, and problem-solving, areas identified as critical to the everyday needs of older adults with lower digital proficiency (Oh et al., 2021). While digital safety and cybersecurity are essential for participation, age-appropriate training remains limited; recent gamified approaches show promise but are rarely adapted to the specific usability constraints of older learners (Abu-Amara et al., 2025; Camilleri et al., 2025). We address these gaps through the development and evaluation of two complementary approaches: **INFINITY**, which integrates traditional board game mechanics with digital question cards to emphasize interaction, and **DiGiUP**, a narrative-driven simulation of real-world digital challenges. Our objectives are to: (a) evaluate the effectiveness of these games in improving and assessing digital competencies; (b) assess associated engagement and enjoyment levels; (c) propose a reusable method for integrated learning and assessment; and (d) identify design refinements for future age-inclusive digital tools.

The study makes four principal contributions to research and practice in digital competence development for older adults. First, it proposes and empirically validates a blended GBA and GBL pathway tailored specifically to older learners, demonstrating how assessment and learning can be integrated within a coherent, low-pressure gameplay environment rather than treated as separate phases. Second, it operationalises the DigComp 2.2 framework directly within concrete game mechanics, providing a replicable design model for embedding competence descriptors at task level and thereby translating a high-level European competence framework into accessible, practice-oriented learning interactions. Third, it contributes theoretically by grounding the intervention in Self-Determination Theory and empirically examining autonomy, competence, and relatedness as motivational drivers of engagement in later-life digital learning. Finally, it offers an ecologically validated implementation in real-world adult education settings, supported by quantitative, non-parametric statistical analyses and qualitative SDT-based coding of trainer observations, thereby providing methodological guidance for future inclusive, game-based digital skills interventions targeting ageing populations.

The paper is structured as follows: Section 2 presents a comprehensive overview of the field, situating our work within the state of the art. Section 3 introduces the methodology of the developed games their design and evaluation principles, as well as reliability and validity instruments, whereas the development and description of the games are provided in Section 4. The results and analysis of the testing and evaluation of the gameplay is presented in Section 5. Section 6 provides a more detailed discussion of the key findings, recommendations, limitations, and future steps. The paper concludes in Section 7.

2. Overview of Related Work

2.1 Digital and interactive learning environments

Digital technology has transformed modern life, creating a pressing need for digital literacy across all demographics, particularly older adults. Digital competence serves not only as a skill for accessing technology but as a key enabler for participation in a digital society. Over the last decade, numerous initiatives have aimed to enhance social inclusion and promote societal involvement among older adults (Camilleri et al., 2025; Casselden, 2023; Oppl & Stary, 2020; Perim et al., 2024). However, research identifies an innate reluctance in this demographic, often rooted in the perception that the digital world is

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the exclusive realm of "digital natives" (Schirmer et al., 2023);L. S. Clark, 2012). Many struggle with the concept of fast-changing knowledge that appears to devalue previously acquired competencies.

In their study, Casselden et al. (2023) emphasizes the need for simple, non-intimidating learning environments to reduce this technology anxiety. In this context, touchscreen-based gaming has proven to be an effective and pleasant medium for digital adoption (Blažič & Blažič, 2020). Central to this success is motivation, which Self-Determination Theory (SDT) frames through the satisfaction of three basic psychological needs: autonomy, competence, and relatedness (Ryan et al., 2006; Ryan & Deci, 2017). Learning environments supporting these needs foster intrinsic motivation and sustained engagement (Tyack & Mekler, 2020). SDT-informed design, offering choice, feedback, and social interaction, can mitigate the fear of failure (Koivisto & Hamari, 2019), However, empirical studies that simultaneously integrate SDT-based motivation with frameworks like DigComp remain scarce.

2.2. Game-based approaches and digital competence frameworks

Game-based approaches (GBL and GBA) are effective in creating safe environments that support incremental, collaborative learning (Alam et al., 2024; Pacheco-Velazquez et al., 2024; Schirmer et al., 2023). Beyond skill acquisition, effective design should integrate emotional signals to improve user engagement and response (de Sousa Borges et al., 2014) (Anolli et al., 2010). Consequently, game-based methodologies often focus on integrating problem-solving, teamwork, and decision-making by offering low-pressure environments for development (Kim & Ifenthaler, 2019; Perim et al., 2024; Xu et al., 2025). Compared to traditional methods, these approaches allow for more nuanced evaluations of cognitive and behavioural skills (A. Pak, 2024; Landers & Sanchez, 2022). Furthermore, games assess higher-order thinking through engaging mechanics (Shute & Ke, 2012) and help overcome technology-related fears through gamified blended learning (Koivisto & Malik, 2021; Kopeć et al., 2017). The common property of all these approaches is the provision of a comfortable atmosphere and training support that enhances older adults' self-confidence, leading to significant improvement in adoption and engagement with digital technologies.

Current design approaches often integrate cognitive skills, emotional intelligence, and user personality (Landers & Sanchez, 2022). Specifically, games serve as effective tools for knowledge, skill, behaviour, and performance assessment (D. B. Clark et al., 2016) (Mislevy et al., 2012). These are especially relevant with the advancement of AI-based (Abbes et al., 2024; Eun et al., 2023) and immersive approaches for learning and education (Buragohain & Chaudhary, 2025). Finally, game-based solutions have also gained popularity in workplace settings, where the use of serious games (designed for a purpose other than for pure entertainment) are being applied as important evaluation tools (al-Qallawi & Raghavan, 2022). Understanding how players react emotionally to game-based learning scenarios remains vital for design.

Despite these advancements, limited studies specifically address older populations, as research remains focused on younger demographics (Gutiérrez-Pérez et al., 2023). Existing work frequently addresses GBL and GBA separately, and many digital-only interventions prove ineffective for those with low proficiency (Perim et al., 2024). While recent digital education research highlights competence-oriented designs where tasks align with framework descriptors (Buragohain et al., 2024; Cleveland-Innes et al., 2024; Constantinou & Rybska, 2024), operationalizing DigComp 2.2 within concrete mechanics remains an underexplored challenge (Furtáková, 2024).

Table 1 synthesizes major works in this field, demonstrating that this study is unique in integrating older adult inclusion, GBL/GBA blending, and SDT-based analysis within a single empirically evaluated approach.

Table 1. Comparative analysis of related work on game-based approaches for digital competence development

Approach	Focus on older adults	Uses GBL	Uses GBA	Aligned with DigComp	SDT-based / motivational analysis	Analogue / hybrid approach	Empirical evaluation
(Blažič & Blažič, 2020)	✓	✓	✗	✗	✗	✓	✓
(Casselden, 2023)	✓	✗	✗	✗	✗	✓	✓
(Abbes et al., 2024)	✗	✓	✓	✗	✗	✗	✓
(Camilleri et al., 2025)	✓	✓	✗	✗	✗	✓	✓
(Schirmer et al., 2023)	✓	✓	✗	✗	✗	✓	✓
(Perim et al., 2024)	✓	✓	✗	✓	✗	✓	✓
(Alam et al., 2024)	✗	✓	✗	✗	✗	✗	✓
(Koivisto & Malik, 2021)	✗	✓	✗	✗	✓	✗	✗
(Pacheco-Velazquez et al., 2024)	✗	✓	✗	✗	✗	✗	✓
(A. Pak, 2024)	✓	✗	✓	✗	✗	✗	✓

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(Kim & Ifenthaler, 2019)	✗	✓	✓	✗	✗	✗	✓
(Landers et al., 2017)	✗	✗	✓	✗	✗	✗	✓
(Shute & Ke, 2012)	✗	✗	✓	✗	✗	✗	✓
(Xu et al., 2025)	✓	✓	✓	✗	✗	✗	✗
(Furtáková, 2024)	✗	✗	✗	✓	✗	✗	✗
(Eun et al., 2023)	✗	✓	✓	✗	✗	✗	✓
This study	✓	✓	✓	✓	✓	✓	✓

This study contributes to addressing the above challenges by developing and evaluating a blended approach to learning and assessment of digital competencies designed explicitly for older adults. By examining the two complementary game-based approaches aligned with DigComp and analysing them through the lens of SDT, we contribute to understanding how interactive learning environments can simultaneously support competence assessment, learning progression, and motivation in lifelong learning contexts. The following section presents the methodology for developing, evaluating, and validating the approach through real-world pilots with older adults.

3. Methodology

This study evaluates a blended game-based approach based on selected competence areas from the "European Framework of Digital Competencies of Citizens" – DigComp 2.2 (Vuorikari et al., 2022). The framework comprises five main digital competencee areas, as shown in

.

Fig. 1: The DigComp 2.2. conceptual reference model

For the development of the two games in this work, three main competence areas were selected from the DigComp framework: *Information and data literacy, Safety, and Problem solving* (see Fig. **2**), as they address critical everyday needs of older adults with lower level of digital proficiency (Oh et al., 2021). Each area contains activities introduced in the two games to enable blended learning and assessment. The game elements follow the MDA (Mechanics, Dynamics, and Aesthetics) design framework (Hunicke et al., 2004), where Mechanics represent the rules and algorithms, Dynamics describe the behaviours manifested during play, and Aesthetics encompass the evoked emotional responses. The full design specifications for both games, together with the description of the gameplay, are available in the study's GitHub repository (Anonymized GitHub, 2026).

Fig. 2: The three DigComp 2.2 competence areas relevant for our work

The resulting games, INFINITY and DiGiUP, integrate assessment and learning within a single pathway. INFINITY is designed to assess baseline competencies while building learner confidence; conversely, DiGiUP evaluates skills acquired during INFINITY while providing further learning through a blend of digital and analogue elements. This iterative process allows for the quantification of digital proficiency levels using the DigComp 2.2 framework. The Erasmus+ project DIGIBLEND (2024) served as the development testbed, ensuring the methodology is evidence-based and replicable.

Both games utilize a familiar board-game format with playing cards to guide participants along a predefined learning path. This analogue-hybrid approach leverages the participants' existing familiarity with card games to reduce technology anxiety and ensure comfort for less-skilled players. Functionally, the games follow a "step-by-step" progression, where players must correctly resolve scenario-based questions to advance. This structure facilitates incremental knowledge acquisition while encouraging the peer-to-peer cooperation necessary for complex tasks.

3.1 Participants and setting

Participant selection and recruitment: Participants were recruited through convenience sampling, which is appropriate for the applied, field-based nature of the study. Trainers introduced the aims of the study and the structure of the gameplay sessions to prospective participants. All participation was voluntary, with written informed consent obtained from all participants prior to participation. No financial or material incentives were provided.

This study was designed as a real-world evaluation within the Erasmus+ DIGIBLEND project testbed. It was conducted in three day-care centres for older adults in Slovenia between April and July 2024. These centres provided a controlled, familiar environment for non-residential daily activities. All sessions were conducted on site, in small groups, and were moderated by trained facilitators. The sample size (N=45) was determined by the logistical constraints of the collaborating centres and the

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small-group nature of the intervention (11 moderated sessions for INFINITY and DiGiUP). This sample size was sufficient to (i) assess usability and acceptance, (ii) observe competence progression patterns across DigComp levels, and (iii) provide stable descriptive and inferential estimates for an exploratory intervention study, while acknowledging limits to generalisability.

Inclusion criteria: Eligibility required participants to be: (i) aged 65 years or older, (ii) capable to engage in moderated table-based gameplay, and (iii) able to provide informed consent (directly and, where applicable, through a legal guardian/caregiver according to the facility's procedures). Exclusion criteria included severe health limitations that would prevent session engagement, as determined by the centre's standard participation rules. However, participants neither met the exclusion criteria nor withdrew during the study, resulting in a 100% completion rate. Consequently, *no cases were excluded from the analysis due to missing or invalid data*.

Participant characteristics and baseline: The final sample comprised of 45 older adults (34 men, 11 women), all aged ≥65 years, who participated in structured gameplay sessions for both INFINITY and DiGiUP. A baseline initial (control) session was used to establish participants' starting digital competence levels, which were concentrated in DigComp Foundation Levels 1-2. Table 2 summarises the demographic characteristics of the participants, including age distribution, educational background, and prior experience with digital devices. These variables were collected to contextualise learning outcomes and to account for heterogeneity in prior exposure to digital technologies.

Table 2. Participant demographics (Note: Percentages may not total 100 due to rounding and overlapping device categories.)

Characteristic	Category	n	%
Age group	65–69	8	17.8
	70–74	14	31.1
	75–79	13	28.9
	80+	10	22.2
Education level	Primary education or less	7	15.6
	Secondary education	13	28.9
	Higher education	25	55.6
Prior device use	Regular smartphone use	19	42.2
	Occasional smartphone use	14	31.1
	No smartphone use	12	26.7
	Regular computer/tablet use	17	37.8

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The study was conducted only in Slovenia for several reasons: first, the institutional scope and funding framework of the Erasmus+ DIGIBLEND project designated Slovenia as a pilot implementation site because it represents a medium-sized EU country with a rapidly ageing population and documented digital inclusion challenges among older adults. This is consistent with trends observed across many European regions. Second, having solid testing results from a relatively monolithic (in cultural sense) environment helps to establish a stronger baseline upon which future tests within more diverse environments would be carried out.

While the sample is not statistically representative of all older adults, we prioritised ecological validity, testing the intervention in real-world educational settings with typical users of adult learning services. The reliance on established frameworks (DigComp 2.2 and Self-Determination Theory), the use of non-language-specific game mechanics, the focus on widely shared digital tasks (e.g., email safety, password practices, device handling), and the availability of all data, protocol and code underlying the statistical analysis support conceptual transferability beyond the Slovenian context. Nevertheless, we acknowledge that external validity is limited by the country-specific setting, convenience sampling, and relatively small sample size.

3.2 Reliability and validity of instruments

In the testing and validation of results, we used three main measurement instruments: (A) DigComp-based performance assessment, (B) Game-Play Questionnaire (SDT-based), and (C) trainer observations/qualitative notes.

Digital competence was operationalised using DigComp 2.2 proficiency descriptors. For both games, each question/task was categorised by DigComp area and competence and aligned to a specific proficiency level with an explicit justification. The mapping from games' questions and tasks to DigComp levels can be found in the GitHub repository (Anonymized GitHub, 2026). This provided content validity by ensuring direct correspondence between gameplay tasks and the recognised competence framework. During sessions, trainers (game masters) moderated gameplay, ensured consistent rule application,

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and verified participants' actions and answers. Correct/incorrect responses were tracked systematically during gameplay to generate participant proficiency profiles and pre-post comparisons. Finally, a baseline control session established initial proficiency distributions prior to the iterative gameplay exposure, supporting interpretability of subsequent shifts in proficiency levels. To strengthen reliability of the DigComp mapping, two authors independently reviewed the task-to-descriptor alignment and resolved discrepancies by consensus; the final mapping table (Table 3) documents the rationale for each assignment.

Furthermore, we measured player experience using a Gameplay Questionnaire (GPQ) grounded in Self-Determination Theory (SDT), capturing autonomy, competence, and relatedness. . In terms of response scale standardisation, participants rated agreement with GPQ statements on a 5-point Likert scale (1=Strongly disagree to 5=Strongly agree). Internal consistency reliability of the GPQ subscales was then assessed using Cronbach's alpha for each GPQ subscale (Autonomy, Competence, Relatedness).

Finally, qualitative data were collected through trainer observations and participant feedback during and after sessions. Notes were coded for recurring themes (e.g., engagement, collaboration, anxiety reduction, and perceived usefulness), providing methodological triangulation with the quantitative outcomes. To improve dependability, a shared coding scheme was used; a second coder reviewed a subset of excerpts and disagreements were resolved through discussion.

The methodological application of each of the described instruments is described in greater detail in the following sections.

3.3 Testing sessions and data analysis

In order to ensure successful teaching of basic digital competences, it was necessary to assess and acknowledge the needs and abilities of the older adults that will play the developed games. The game designers were aware that digital competencies vary among the members of any group of older adults; consequently, the first challenge in the game development was to determine the actual level of digital competence of the target group members. Besides the information about the age and the existence of some competences, other game elements were explored, like the selection of the type of the game, the incentives to be given to the players, the difficulty levels applied and the roles of the players. Important issues that were explored were the game mechanics (rules, challenges, competitions, rewards) and dynamics (narratives, storylines). Specification about the real-time actions allowed to the learners and the behaviours stemming from players' key decisions during the game-specific events were specified and collected. The game aesthetics was also critical, and its evolution was also observed during and after the play.

The study addressed the following research questions:

- **RQ1:** How can game-based approaches support the assessment and development of digital competences among older adults?
- **RQ2:** What is the added value of integrating game-based assessment and game-based learning?
- **RQ3:** How do motivational factors influence engagement with game-based digital skills training?

First, trainers informed prospective participants of the purpose and structure of testing sessions. We then obtained consent for participation from each of them, including ethical and privacy statements for the used methods, the processing of personal data, and the publishing of the processed data and results. A template of the consent form used is provided in **part A1 of the Supplementary Information**.

Data were collected during game sessions and through post-game evaluations, incorporating both observational and self-reported methods. Participants engaged in structured gameplay sessions moderated by the trainers (as game masters), who ensured smooth game flow and verified participants' actions and answers. They also observed participant behaviour, noting levels of engagement, interaction, and progression through the games. The INFINITY game was played in 11 sessions, each involving four participants and lasting approximately 50-55 minutes per session. The DiGiUP game followed a comparable small-group format, with sessions lasting approximately 60-70 minutes, reflecting the more narrative-driven structure and scenario-based progression. Each session was conducted as a single, self-contained activity, including a brief introduction by the trainers, the gameplay phase, and a short reflective discussion at the end. Participants completed an initial baseline assessment session prior to gameplay and a post-game evaluation immediately following the completion of each game. No repeated sessions were required from the same participant beyond the planned gameplay exposure, ensuring consistency across testing conditions.

For quantitative assessment of digital competencies, we employed the DigComp 2.2 framework to determine the proficiency levels, according to the systematisation shown in Fig. **3**.

Fig. 3: The DigComp 2.2. proficiency levels

We focus on the three areas of digital competence: *Information and data literacy*, *Safety*, and *Problem solving*. Each area includes specific competences, and each competence is broken down into 3 overall proficiency levels, following the DigComp framework: Foundation (Level 1), Intermediate (Level 2), and Advanced (Level 3). We categorized each question and task according to the DigComp competence it addresses and aligned its difficulty with a specific proficiency level descriptor. Table 3 shows an example of this mapping, whereas the complete set of questions and their systematization by levels can be found in the GitHub repository (Anonymized GitHub, 2026).

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Table 3. Mapping questions/tasks to DigComp 2.2 proficiency levels

Game Task	DigComp Area	Competence	Proficiency Level	Justification
Identify a secure password	Safety	Protecting devices and content	Level 2	Recognizes safe password practices with support
Evaluate an email for phishing	Safety	Protecting personal data and privacy	Level 3	Can recognize suspicious content and knows basic prevention strategies
Search for information online	Information and data literacy	Browsing, searching and filtering data	Level 1	Can conduct basic searches with some assistance
Download and install a mobile app	Problem solving	Solving technical problems	Level 3	Can complete straightforward installation tasks independently
Use QR codes to access learning material	Problem solving	Identifying needs and technological responses	Level 3	Can select appropriate tools to access information
…	…	…	…	…

Following the mapping of tasks and questions, we tracked which level-specific tasks the players succeed at during gameplay, which allowed us to create a proficiency profile for each player. An example of this is shown in Table 4, whereas the complete proficiency profiles for all players can be found in the paper's GitHub repository (Anonymized GitHub, 2026). This enabled us to determine the number of participants that have advanced their proficiency level in a specific area and, moreover, the progression between the first (INFINITY) and the second game (DiGiUP).

Table 4. Proficiency profile per player

Participant	Info & data literacy	Safety	Problem solving
1	Level 1	Level 2	Level 2
2	Level 2	Level 1	Level 2
3	Level 1	Level 1	Level 2
4	Level 3	Level 3	Level 3
5	Level 2	Level 1	Level 2
6	Level 1	Level 2	Level 1
…	…	…	…

In addition to the quantitative analysis, trainers provided qualitative feedback, including anecdotal evidence on how participants adapted to the games and overcame initial hesitations. Trainers' notes and participant feedback were coded and analysed according to the Self-determination theory, i.e., for autonomy, competence, and relatedness. Their observations were critical in evaluating the games' ability to reduce participants' anxiety and encourage collaboration. Moreover, they noted down the participants' statements and reactions to the games, during and post-gameplay. Specific anecdotes highlighted how personalized guidance and familiar game formats helped alleviate participants' initial apprehensions. The trainers' notes, together with their coding, are openly available on GitHub (Anonymized GitHub, 2026).

Player experience and motivational aspects of the game-based intervention were measured using a self-developed Gameplay Questionnaire (GPQ) grounded in Self-Determination Theory (SDT) (Ryan et al., 2006). SDT posits that motivation and engagement are driven by the satisfaction of three basic psychological needs: *autonomy, competence*, and *relatedness*. *Autonomy* implies that older adults can make choices within the game and adjust their experience to their comfort level. *Competence* means that there are opportunities to develop and demonstrate mastery of tasks during the game play, whereas *relatedness* signifies that a game can foster collaboration, which, in turn reduces the feeling of social isolation while promoting mutual learning. These constructs are widely used and theoretically validated in studies of learning, games, and technology adoption, supporting the construct validity of the instrument (F. Gao, 2024; Salikhova et al., 2020).

The GPQ was designed to operationalise these three SDT dimensions in the specific context of older adults engaging with game-based digital skills training. Questionnaire items were formulated to be short, concrete, and accessible, avoiding technical

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terminology and abstract phrasing, in order to ensure comprehension by participants with varying educational backgrounds and limited digital experience. The full list of questionnaire items is provided in **part A2 of the Supplementary Information**. The GPQ consisted of 10 items: 3 items were specific to *Autonomy*, 4 items captured the *Competence*-related aspect, and 3 were assigned to *Relatedness*. Content validity was ensured through an iterative development process in which items were reviewed by the research team and facilitators experienced in adult education and digital literacy training. Participants responded to each item using a 5-point Likert scale ranging from 1 ("Strongly disagree") to 5 ("Strongly agree"). Internal consistency reliability was assessed using Cronbach's alpha, with detailed results reported in Section 5.3. The data, code, and all calculations are openly available in the GitHub repository (Anonymized GitHub, 2026).

Confirmatory factor analysis (CFA) was not performed in this study due to the exploratory and pilot nature of the research, the context-specific adaptation of the questionnaire, and the sample size. The primary aim of the GPQ was to obtain an initial, theory-informed assessment of motivational experiences among older adults in a real-world educational setting, rather than to establish a fully generalisable psychometric instrument. Given the relatively small sample (N = 45), conducting CFA would not meet commonly recommended sample size requirements for stable parameter estimation. Instead, construct validity was supported through the theoretical grounding of the instrument in Self-Determination Theory, careful item design, and the assessment of internal consistency reliability using Cronbach's alpha. Future studies with larger and more diverse samples will allow for more advanced psychometric validation, including exploratory and confirmatory factor analyses.

4. Game Development and Description

In this section, we briefly describe the design of INFINITY and DiGiUP and provide a mapping of their elements onto the MDA framework. The detailed specifications and gameplay rules and dynamics can be found in the corresponding file in the GitHub repository (Anonymized GitHub, 2026).

4.1 The INFINITY game

INFINITY offers a blend of learning, competition, and strategic decision-making. By navigating challenges, leveraging special fields, and answering questions of varying difficulty, players engage in an immersive experience that enhances their digital skills while fostering a sense of accomplishment and camaraderie. It operates on a game-board shaped like the mathematical infinity symbol (∞), measuring 75 × 60 cm, as presented in Fig. **4**. The board is punctuated with special fields that add complexity and strategy to gameplay.

Fig. 4: The INFINITY game-board design

The game begins with each player selecting a game totem, choosing one of the offered colors (red, blue, green, or yellow) to represent their identity on the board. It unfolds as players take turns rolling the die, which allows them to advance their totems along the infinity-shaped path. The objective is to conquer eight fields on the board by answering questions that test digital competencies. As players move through the fields, they face various scenarios that shape the dynamics of their gameplay and lead to opening cards from various card decks (**Error! Reference source not found.**). The game ends when one player successfully conquers eight fields. However, the overall winner is determined not only by the number of conquered fields, but also by the number of correctly answered questions.

Fig. 5: The INFINITY cards with the quiz-type questions

The dynamics of the game focus on strategy, competition, and skill assessment. In terms of aesthetics, the game aims to create an engaging, competitive, and educational experience. Table 5 presents the description of the game by mapping its elements onto the MDA framework.

Table 5. The INFINITY game mapped onto the MDA framework

MDA Framework	**INFINITY Game elements**	
Mechanics	*Game-board fields*	The board consists of 42 active nodes, that regulate player movement and tactical advantages.
	Card decks	Four distinct card decks that provide standardized assessment tasks categorized by difficulty alongside strategic Joker cards that allow players to bypass specific challenges.

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	Player Totems & Dice	Movement along the infinity-shaped path is governed by a randomized six-sided die roll, with four color-coded totems representing individual player identities and progress.
	Winning conditions	The first player to conquer 8 fields ends the game. Players are evaluated based on the number of correctly answered questions.
Dynamics	*Strategic Movement*	Players navigate the infinity-shaped path, aiming to conquer 8 fields by landing on white fields or attacking opponents' conquered fields.
	Question Challenges	Conquering fields involves answering questions from the appropriate card decks. Correct answers allow players to claim or reclaim fields, while incorrect answers delay progress.
	Penalty & Reward Mechanisms	Landing on Debug Zone fields imposes penalties by requiring players to answer advanced questions. Joker cards and Wi-Fi fields provide rewards by offering advantages, such as skipping questions or reclaiming fields without answering.
	Interactive Opportunities	Players can attack conquered fields, creating opportunities for dynamic interactions and competition among participants.
	Skill Assessment	Each correctly or incorrectly answered question is tracked, providing insights into players' digital competencies.
Aesthetics	*Challenge and Mastery*	Players are motivated by the progression of difficulty in the question decks, which test and improve their digital skills.
	Competition & Interaction	Confrontations between players, through field attacks, foster a sense of rivalry and excitement.
	Flexibility & Inclusivity	The difficulty level of the questions can be adjusted to suit different player skill levels, making the game accessible to a wide audience.
	Exploration & Strategy	The infinity-shaped board design and diverse field types encourage players to explore strategic moves and make calculated decisions.
	Rewarding Learning	The game provides an entertaining method for evaluating and improving digital competencies while rewarding correct answers with tangible progress in the game.

Fig. 4Error! Reference source not found.6

4.2 The DiGiUP game

DiGiUP is primarily designed to assess digital competencies through an interactive and narrative-driven gameplay experience, although learning is inherent to the process. The dynamics focus on fostering collaboration, learning, and competition. The game delivers an engaging, educational, and collaborative experience, providing players with meaningful learning experience on topics related to the three digital competences areas of the DigComp 2.2. framework: information and data literacy, safety and problem solving.

In DiGiUP, players take on the role of trusted community members participating in a digital discovery challenge. The game unfolds in a house represented on the game-board, featuring six distinct rooms, as shown in **Error! Reference source not found.**: the Kitchen, Bathroom, Dining Room, Living Room, Bedroom, and Study Room.

Fig. 7: The DiGiUP game board design, a house with 6 rooms: Kitchen, Bathroom, Dining room, Living room, Bedroom and Study room

Gameplay starts by introducing the players to the digital discovery challenge in which they take on the role of trusted community members: "*The local community centre has launched a month-long initiative to promote smart, safe, and confident use of digital tools for everyday tasks, from staying connected with loved ones to managing appointments, online shopping, and ensuring digital safety."* Players move through familiar settings like the kitchen, living room, and study, solving practical digital challenges tailored to real-life situations. Their mission is to complete tasks, gain confidence, and unlock digital milestones that support independence and lifelong learning.

Each of the six rooms shown in **Error! Reference source not found.** corresponds to a specific set of digital competencies from the DigComp 2.2 framework. For instance, the Kitchen and Living Room focus on information and data literacy, and the Bathroom and Bedroom emphasize safety skills. The game's competitive edge comes to life as players strive to reach the final room. The first player to complete all six rooms and solve the last digital task wins.

To provide a systematic outlook of the game elements and the learning process, Table 6 maps the DiGiUP game constituents onto the MDA framework.

Table 6. The DiGiUP game mapped onto the MDA framework

MDA Framework	DiGiUP Game elements	
Mechanics	*Game-board design*	A vertical "house" architecture featuring six thematic rooms where progression is gated by alternating between standard question fields and mandatory QR-triggered digital tasks.
	Player components	Color-coded totems, supported by a resource-management system consisting of help tiles for peer-assistance and bridging cards for accessing supplementary learning modules.
	Question system	A tiered assessment structure utilizing basic, intermediate, and advanced tasks managed via a centralized question book, that maps gameplay content to corresponding DigComp 2.2 areas.
	Winning conditions	The game ends when a player successfully completes the sixth room, with victory determined by a combination of vertical progression speed and the strategic retention of bridging cards as a proxy for mastery.
Dynamics	*Progression through Levels*	Players answer questions to advance through each room. At the end of a room, they face a mandatory digital task at the elevator to move to the next level.
	Strategic use of Help Tiles and Bridging cards	Players can use help tiles to request assistance from others or exchange them for bridging cards, which in turn provide access to additional learning modules.
	Collaborative problem-solving	Help tiles promote teamwork by encouraging players to assist each other in answering questions or solving digital tasks.
	Player-driven difficulty	Players can choose the level of difficulty for each question, allowing them to tailor challenges to their skill level.
	Competition for resources and progress	Limited help tiles and the scoring system for bridging cards introduce strategic decision-making.
Aesthetics	*Narrative immersion*	The story revolves around proving digital competency to take care of grandchildren, providing players with a relatable and motivating context. The narrative creates a sense of purpose as players navigate the rooms and challenges of the house.
	Assessment through challenge and mastery	Players face increasing difficulty through the progression of questions and digital tasks, fostering a sense of accomplishment as they overcome each level.
	Collaborative learning	The integration of help-tiles and bridging cards encourages teamwork and shared learning, making the experience interactive and supportive.
	Customizability and inclusivity	The game accommodates varying skill levels through adjustable question difficulties, ensuring accessibility for diverse participants.
	Rewarding learning	The bridging cards transform incorrect answers into opportunities for learning, reinforcing educational goals.
	Competition and strategy	The race to the top of the house creates a sense of excitement and rivalry, while resource management adds a layer of strategic depth.

5. **Error! Reference source not found.**Results and Analysis

In this section, we report and analyse on the results from the pilot study of the two game-based approaches. First, we overview game-specific results, and then we elaborate on the overall game-based approach in view of the specific contributions and research questions.

5.1 Results of the INFINITY game

The setup for INFINITY in the daycare center is shown in Fig. **8Error! Reference source not found.**. Before starting, all players were introduced to the game's rules and objectives. Each session included four participants, and on average, a session lasted approximately 50-55 minutes. During the game, players successfully conquered an average of 5.5 fields, answering 76% of the questions correctly, while 24% of the questions were answered incorrectly.

Fig. 8: The testing session of the INFINITY game

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After the initial (control) session, all participants were assessed for proficiency according to the DigComp 2.2 framework. The results revealed that most participants fell within the foundation levels (1-2)**,** indicating relatively low levels of digital competency and poor exposure to digital environments. It is important to note that the goal of the assessment was to merely identify the starting competency baselines, and not to evaluate advanced mastery. The chart on Fig. **9** shows the distribution of proficiency levels of participants per competence area.

Fig. 9: INFINITY proficiency level distribution by competence area

Across all three competence areas, participant performance is strongly concentrated at lower proficiency levels, particularly Level 1. Information and Data Literacy show the highest frequency at Level 1, indicating that the majority of participants demonstrated only foundational competences in this area. A smaller proportion reached Level 2, while only a minimal number attained Level 3. A similar pattern is observed for Safety, although the distribution is slightly more balanced between Levels 1 and 2. While Level 1 remains the most frequent, Safety exhibits a comparatively higher presence at Level 2 than the other competence areas, suggesting marginally stronger foundational awareness. Nevertheless, progression beyond Level 2 is limited, with only a small number of participants reaching Level 3. For Problem Solving, the distribution is even more skewed toward Level 1, indicating substantial difficulty in demonstrating higher-order digital problem-solving skills within the INFINITY assessment. Only a limited subset of participants reached Level 2 or Level 3. From these results, it can be concluded that the INFINITY proficiency distributions reveal a pronounced floor effect, with performance clustering at Foundation levels (Levels 1-2) across all competence areas. This distribution provides a useful baseline against which the more differentiated proficiency profiles observed with DiGiUP can be interpreted.

5.2 Results of the DiGiUP game

DiGiUP provided an engaging and interactive experience, yielding valuable insights into players' acquired skills, as well as on its impact on the participants experience (see Fig. **10Error! Reference source not found.**). Trainers, who were responsible for facilitating the game's progression, noted that the gameplay effectively stimulated engagement and cooperation among players. This was accompanied by an enhanced sense of curiosity, as participants expressed excitement about discovering "what's next" during each stage of the game, demonstrating their competencies without the sense of being evaluated or scrutinized. Players particularly appreciated the dynamic elements of the game, including the digital challenges and tasks, as well as the incorporation of peer-learning experiences that fostered collaboration and communication.

Fig. 10: The testing session of the DiGiUP game

While an average session lasting approximately 14 minutes longer than sessions of INFINITY, DiGiUP's participants appeared more comfortable with the extended duration. The longer gameplay was well-received, as it encouraged ongoing discussions and meaningful interactions among players, making the experience more immersive and engaging.

All participants had previously played 10 sessions of the INFINITY game. Hence, this was their second exposure to digital learning through gameplay, but with similar knowledge and skill requirements. For better understanding of the improvement in digital competencies, Fig. **11** illustrates the distribution of DigComp proficiency levels achieved by participants when assessed using DiGiUP, across the competence areas, with INFINITY shown for baseline comparison.

Fig. 11: Comparison of proficiency levels: INFINITY vs. DiGiUP

For Information and Data Literacy, the DiGiUP results show a clear upward redistribution of proficiency levels. Approximately 84% of participants (Levels 2 and 3 combined) achieved intermediate or advanced proficiency, while only 16% remained at Level 1, compared to 62% at Level 1 in the INFINITY baseline. Nearly half of the participants (49%) reached Level 3, indicating that DiGiUP facilitated progression beyond foundational recognition skills toward more independent information handling and evaluation. In the Safety competence area, DiGiUP demonstrated a similarly pronounced shift. Only 13% of participants remained at Level 1, while 87% reached Levels 2 and 3 combined, with the majority (56%) achieving Level 3. Compared to the INFINITY distribution, which was concentrated primarily at Levels 1 and 2, this redistribution suggests that DiGiUP more effectively elicits applied digital safety competences, including risk identification and appropriate protective responses. The effect of DiGiUP was particularly evident in Problem Solving. The proportion of participants at Level 1 decreased substantially to 11%, while 89% reach Levels 2 and 3, with over one-third (38%) attaining Level 3. In contrast, the INFINITY baseline was strongly skewed toward Level 1 (64%). This marked redistribution underscores DiGiUP's capacity to support more complex digital reasoning and task-based problem-solving behaviours.

The DiGiUP proficiency distributions show a substantial reduction of floor effects and a consistent redistribution toward intermediate and advanced DigComp levels across all competence areas. These patterns indicate that DiGiUP provides a more

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discriminative and informative assessment of digital competence among older adults, complementing the baseline-oriented insights obtained with INFINITY.

5.3 Statistical analysis

5.3.1 Digital competence per DigComp area

To compare participants' demonstrated digital competence levels across the two game-based approaches (INFINITY and DiGiUP) and to quantify the magnitude of observed differences, statistical analyses were conducted for each competence area. This analysis is shown in Fig. **12**. The results reveal that across all DigComp areas, DiGiUP yielded systematically higher proficiency levels than INFINITY. Mean DigComp levels increased by 0.73 to approximately one full level, i.e.: $\Delta = +0.96$ or +66.2% for Information and Data Literacy, $\Delta = +0.78$ or +51.5% for Safety, and $\Delta = + 0.73$ or +42.3% for Problem Solving. While INFINITY results were concentrated at Foundation levels (Levels 1-2), DiGiUP distributions shifted toward Intermediate levels (Levels 2-3) and even reached Advanced level (Level 3), with higher medians and increased variance. These patterns indicate that DiGiUP elicited a broader and higher range of demonstrated competences across all assessed areas.

Fig. 12. Mean proficiency levels per competence area (INFINITY vs. DiGiUP)

To further assess differences between INFINITY and DiGiUP, we employed Wilcoxon signed-rank tests (the code and data are available in the GitHub repository). This non-parametric test was selected because: (i) the DigComp levels are ordinal rather than interval-scaled; (ii) measurements are paired, as each participant completed both assessments; and (iii) normality assumptions required for parametric tests (e.g., paired t-tests) could not be assumed, particularly given the observed floor effects for INFINITY. Statistical significance was assessed using a two-tailed criterion with $\alpha = 0.05$. Effect size values were interpreted using conventional benchmarks ($r \approx 0.10$ small, $r \approx 0.30$ medium, $r \geq 0.50$ large). Reporting effect sizes were considered essential given the exploratory nature of the study and the limitations of p-values in conveying substantive importance. Across all competence areas, DiGiUP yielded significantly higher results (all $p < 0.001$). Effect size estimates indicated very large effects ($r \approx 0.98$–1.05), indicating a highly consistent and systematic shift toward higher proficiency levels under DiGiUP. In practical terms, this means that virtually every participant demonstrated higher proficiency under DiGiUP than under INFINITY, across all three competence areas. In all three areas, the Wilcoxon statistic equalled $W = 0$, reflecting the absence of negative ranks, which implies that no participant scored higher on INFINITY than on DiGiUP in any competence area. However, it is important to note that, because the study lacks a control group and uses repeated exposure, these results indicate strong associative evidence within the study context rather than definitive causal proof.

5.3.2 Game-Play Questionnaire data analysis

Following each session, participants completed the SDT-based Game-Play Questionnaire (see for the questionnaire's template in **part A2 of the Supplementary Information**) to assess their perceptions and learning outcomes across three dimensions: *Autonomy*, *Competence*, and *Relatedness*. Fig. **13** shows the results for each dimension, with the scores given per item.

The Autonomy subscale consists of three items: *A1: felt free to make my own choices while playing the games; A2: felt that I could play the games in my own way; and A3: felt in control of my actions during the games.* As the figure shows, responses are concentrated in the upper range (3-5), with A2 and A3 showing high similarity, while A1 demonstrating slightly more variability. Cronbach's alpha for the autonomy scale α=0.78, indicating acceptable internal consistency and reliable measurement across the three autonomy-related items.

Fig. **13Error! Reference source not found.**a) presents the distribution of responses to the three *Autonomy* items measured on a 5-point Likert scale. Overall, responses are strongly skewed toward the upper end of the scale, indicating high perceived autonomy during gameplay. For Item 1, approximately 91% of participants selected "5" (Strongly Agree), while the remaining 9% selected "4"; no responses were recorded below level 4. Item 2 shows greater dispersion, with about 22% of participants selecting "3", 58% selecting "4", and 20% selecting "5". Item 3 follows a similar pattern, with approximately 13% of responses at "3", 67% at "4", and 18% at "5". Across all items, no responses were observed at levels 1 or 2, and only a minority of participants selected the neutral midpoint. This concentration at higher response categories, combined with modest variation across items, explains the acceptable but non-perfect internal consistency of the Autonomy subscale (Cronbach's $\alpha = 0.83$), indicating that the items measure closely related yet complementary aspects of perceived autonomy. From a Self-Determination Theory perspective, these results suggest that the game-based activities supported participants' basic need for autonomy, which is a key driver of intrinsic motivation and sustained engagement in learning.

Fig. 13: Results from the Game-Play Questionnaire

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The Competence subscale consists of four items: *C1: felt capable of completing tasks; C2: felt more confident about digital skills; C3: felt successful during gameplay;* and *C4: perceived task difficulty as appropriate.* Unlike the Autonomy, there is a visible variation between items, particularly at lower score ranges (2-3). The calculated Cronbach's alpha for the Competence subscale is α=0.95, indicating good internal consistency and reliable measurement of perceived competence across the four questionnaire items.

Fig. **13**b) presents the distribution of responses to the four Competence items measured on a 5-point Likert scale. Across all items, responses are concentrated at the upper end of the scale, indicating generally high perceived competence during gameplay. For Item 1, approximately 67% of participants selected "4", while about 16% selected "5" and 13% selected "3"; only a small proportion (≈4%) selected "2", with no responses at level 1. Item 2 shows a similar pattern, with around 62% of responses at "4", 24% at "3", and 13% at "5". Item 3 exhibits slightly greater dispersion, with about 53% selecting "4", 36% selecting "3", and 9% selecting "5". For Item 4, responses again cluster at higher levels, with approximately 60% of participants selecting "4", 18% selecting "5", and 18% selecting "3", and only a very small proportion at "2". Across all four items, no responses were recorded at level 1, and responses at level 2 were rare. This pattern reflects consistently positive competence perceptions with moderate item-level variation, which aligns with the good internal consistency of the Competence subscale (Cronbach's α = 0.95). From a Self-Determination Theory perspective, these results indicate that the game-based activities supported participants' sense of competence, a core psychological need that underpins intrinsic interest and confidence in learning new digital skills.

Finally, the Relatedness subscale consists of three items: *R1: enjoyed interacting with other participants; R2: felt supported by the trainer and other players;* and *R3: felt connected to others during gameplay*. It demonstrated acceptable internal consistency (Cronbach's α=0.73), indicating reliable measurement despite a restricted response range at the upper end of the scale. This is appropriate for exploratory and applied research with small to moderate samples as ours. The value reflects strong perceived relatedness with a mild ceiling effect, but without item redundancy, which is aligned with older-adult populations questionnaires with restricted response ranges.

Fig. **13**c) presents the distribution of responses to the three Relatedness items measured on a 5-point Likert scale. Across all items, responses are strongly concentrated at the highest end of the scale, indicating very high perceived social connectedness during gameplay. For Item 1, approximately 76% of participants selected "5", while the remaining 24% selected "4"; no responses were recorded below level 4. Item 2 shows an even stronger concentration at the top of the scale, with around 91% of participants selecting "5" and 9% selecting "4", and no responses at levels 1–3. Item 3 displays slightly more dispersion, with about 40% of participants selecting "5", 56% selecting "4", and a small minority (≈4%) selecting "3". Across all items, responses at levels 1 and 2 are absent, and only a negligible proportion of responses fall at the neutral midpoint. However, the markedly lower variance of Item 2 compared with Item 3 reduces inter-item covariance, which explains the acceptable (but not high) internal consistency of the Relatedness subscale (Cronbach's α = 0.73). From a Self-Determination Theory perspective, these results suggest that the game-based activities strongly supported participants' need for relatedness, while the differential response patterns across items reflect nuanced aspects of social connectedness rather than item redundancy.

Table 7 presents the statistics related to each psychological need (construct) as defined by STD, showing the item variance and the corrected item-total correlation (required for computing the Cronbach's alpha), as well as the mean, standard deviation, and the min and max values per each GPQ item.

All data and computations based on these data are openly available in the study's GitHub repository (Anonymized GitHub, 2026).

Table 7. Statistics per item for all STD constructs

Items	Item Variance	Corrected Item–Total Correlation	Mean	Std.	Min	Max
Autonomy [Cronbach's alpha (α): 0.83]						
A1. I felt free to make my own choices while playing the games.	0.08283	0.52551	4.91111	0.28780	4.0	5.0
A2. I felt that I could play the games in my own way.	0.43131	0.89842	3.97778	0.65674	3.0	5.0
A3. I felt in control of my actions during the games.	0.33636	0.92049	4.06667	0.57997	3.0	5.0
Competence [Cronbach's alpha (α): 0.95]						
C1. I felt capable of completing the tasks in the games.	0.47273	0.85926	3.93333	0.68755	2.0	5.0
C2. The games helped me feel more confident about my digital skills.	0.37374	0.89344	3.88889	0.61134	3.0	5.0

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C3. I felt successful when progressing through the games' activities.	0.44647	0.79346	3.68889	0.66818	2.0	5.0
C4. The difficulty of the games' tasks was appropriate for me.	0.53737	0.89992	3.91111	0.73306	2.0	5.0
Relatedness [Cronbach's alpha (α): 0.73]						
R1. I enjoyed interacting with other participants during the games.	0.18889	0.58216	4.75556	0.43461	4.0	5.0
R2. I felt supported by the trainer and other players.	0.08283	0.47978	4.91111	0.28780	4.0	5.0
R3. Playing the games made me feel connected to others.	0.32525	0.53778	4.35556	0.57031	3.0	5.0

Given the ordinal nature of Likert-scale data and the moderate sample size, non-parametric statistical methods were used throughout the analysis. Normality was not assumed, and inferential comparisons were conducted using Wilcoxon signed-rank tests. However, normality was assessed using the Shapiro-Wilk test for descriptive purposes only. The Shapiro-Wilk tests indicated that none of the three SDT subscale scores follow a normal distribution (Autonomy: $W = 0.819$, $p < 0.001$; Competence: $W = 0.903$, $p = 0.001$; Relatedness: $W = 0.809$, $p < 0.001$). This justifies the use of non-parametric statistical tests for subsequent analyses. Thus, differences between SDT constructs were examined using the Friedman test, whereas effect sizes were calculated using Cliff's delta.

The Friedman test revealed a statistically significant difference among the three SDT constructs ($\chi^2 = 63.939$, $p < 0.001$). This indicates that participants' perceived levels of autonomy, competence, and relatedness differ systematically, rather than being uniformly experienced, suggesting differentiated support for specific psychological needs. To measure effect sizes we used Cliff's delta, a non-parametric test indicating the probability that a randomly selected score from one group exceeds a score from another, with values above 0.47 conventionally interpreted as large. The analysis revealed large effect sizes across all pairwise comparisons of the SDT constructs, indicating a systematic motivational profile rather than uniform need satisfaction. Competence was rated significantly higher than both autonomy ($\delta = -0.615$) and relatedness ($\delta = 0.509$), while relatedness was also rated higher than autonomy ($\delta = -0.474$). This pattern implies that the game-based interventions most strongly supported participants' sense of mastery and effectiveness, followed by social connectedness, with perceived autonomy comparatively lower. From an SDT perspective, these results reflect the highly structured and facilitated nature of the gameplay, which prioritised task success, feedback, and collaborative interaction, while offering more limited opportunities for independent choice and self-direction.

5.3.3 Qualitative analysis and anecdotal evidence

The quantitative results are further supported by anecdotal evidence from the trainers' notes. Trainers' observations and anecdotal notes were analysed using a deductive thematic coding approach informed by Self-Determination Theory, following (Naeem et al., 2023) and (Ahmed et al., 2025). The three SDT-constructs, *Autonomy*, *Competence*, and *Relatedness*, were used as top-level codes, with inductively derived sub-codes capturing recurring patterns observed during gameplay sessions. An excerpt of this exercise is provided in Table 8, whereas the complete coding framework with all coded trainers' notes are provided in the GitHub repository (Anonymized GitHub, 2026).

Table 8. SDT-based coding scheme for analysis of trainers' observations

Code	Definition	Example quote (from trainers' notes)
A1: Initiative and choice	Instances where participants voluntarily chose to attempt tasks independently or expressed a sense of control over gameplay decisions.	*"Several participants insisted on trying the task themselves before asking for help, saying they wanted to 'see if they could manage it alone.'"*
A2: Reduced fear of errors	Evidence that participants perceived the game as a safe space where mistakes were acceptable and part of learning.	*"Participants laughed about wrong answers and commented that it felt more like playing than being tested."*
A3: Perceived relevance	Moments where participants connected game tasks to their own everyday digital experiences.	*"One participant related the phishing card directly to an email she had received recently, supposedly sent by her bank."*
C1: Initial low self-efficacy	Expressions of doubt, anxiety, or lack of confidence regarding digital skills at the start of gameplay.	*"At the beginning, several participants stated that they were 'too old for digital things' and expected to fail."*

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C2: Skill acquisition	Observable improvement in task performance or strategy use during gameplay.	*"As the game progressed, participants began identifying suspicious messages more quickly and with fewer prompts."*
C3: Confidence gains	Verbal or behavioural indications of increased confidence or pride in newly acquired skills.	*"Participants expressed surprise at their own abilities, saying things like 'I didn't know I could do this.'"*
R1: Peer support	Instances of participants helping, explaining, or encouraging each other.	*"Participants frequently leaned toward one another to explain answers or clarify rules."*
R2: Shared emotional experience	Collective emotional responses such as humour, laughter, or shared frustration.	*"Mistakes often triggered laughter rather than embarrassment, creating a relaxed group atmosphere."*
R3: Collaborative problem solving	Evidence of joint decision-making, role distribution, or collective strategies.	*"Groups began assigning informal roles, such as one person reading cards while another checked answers."*

Across both games, the trainers' observations indicate that motivational engagement evolved over time, moving from initial hesitation toward active participation and collaboration. INFINITY appeared particularly effective in establishing relatedness and emotional safety through familiar board-game mechanics and humour, which in turn supported gradual confidence building. DiGiUP, by contrast, more strongly supported autonomy and competence, encouraging independent problem solving within meaningful, scenario-based contexts. Importantly, the notes reveal that satisfaction of the three SDT needs was interdependent. Peer support (relatedness) often enabled participants to attempt tasks independently (autonomy), which in turn led to successful task completion and confidence gains (competence). This dynamic interplay aligns closely with SDT predictions and helps explain the high engagement and positive experiential outcomes observed in the quantitative results.

5.4 Key takeaways

The evaluation of the two game-based approaches, INFINITY and DiGiUP, yields several context-bound insights into the design and use of game-based learning and assessment for digital competence development among older adults within the specific pilot setting of this study. First, the results demonstrate that game-based approaches are both feasible and effective for assessing and supporting digital competence development in older adult populations. INFINITY successfully served as a baseline-oriented assessment and confidence-building tool, revealing participants' initial competence profiles, which were largely concentrated at DigComp Foundation levels. In contrast, DiGiUP enabled a clear progression toward intermediate and advanced proficiency levels, particularly in problem solving and information and data literacy. These findings indicate substantial within-sample improvements; however, they should be interpreted as exploratory and context-specific rather than broadly generalisable.

Second, the findings suggest that combining GBA and GBL into a blended pathway may offer practical advantages within similar facilitated adult-learning environments. Rather than treating assessment and learning as separate phases, the study indicates that assessment activities embedded in gameplay can simultaneously function as learning experiences. In this pilot implementation, such integration appeared to reduce test anxiety and support gradual skill development by framing competence evaluation as a supportive rather than judgmental process. These observations are limited to the studied context and require replication in other institutional and cultural settings before broader applicability can be assumed.

Third, the GPQ results grounded in Self-Determination Theory indicate that the games effectively supported the three basic psychological needs of autonomy, competence, and relatedness, which are central to sustained motivation. High scores across all three dimensions suggest that participants felt capable, socially supported and reasonably in control during gameplay. Importantly, acceptable to good internal consistency of the GPQ subscales supports the reliability of these measurements in this specific implementation. Nevertheless, given the small convenience sample and single-country context, these motivational findings should be viewed as indicative rather than definitive.

Finally, the qualitative observations and trainer feedback reinforce the quantitative findings by underscoring the importance of collaboration, familiarity, and facilitation. Familiar board-game metaphors, small-group formats, and trainer moderation appeared to lower barriers to participation among these participants. It is important to note, however, that the observed improvements should be interpreted in light of the study's exploratory design and contextual constrain, as they may reflect not only the game-based approaches, but also repeated task exposure and trainer support during gameplay. Moreover, the Slovenian context may differ from other national and cultural contexts, which implies that the results should be taken as indicative rather than causal.

The key takeaway from our study is that carefully designed game-based approaches, grounded in competence frameworks and motivational theory, may provide an inclusive, engaging, and methodologically sound pathway for digital skills learning and assessment among older adults. The complementary roles of INFINITY and DiGiUP illustrate how different game designs

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can be strategically combined to support confidence-building, learning progression, and meaningful competence evaluation in lifelong learning contexts.

6. Discussion

6.1 Overview of contributions

This study makes several interrelated context-specific contributions to research and practice in game-based learning (GBL), game-based assessment (GBA), and digital competence development for older adults, while directly addressing the research questions posed in Section 4.3.

In response to **RQ1** *(How can game-based approaches support the assessment and development of digital competences among older adults?)*, the results demonstrate that both INFINITY and DiGiUP were feasible and usable tools within this adult-education setting. INFINITY primarily supported baseline assessment and confidence building, revealing participants' initial competence profiles, which were predominantly situated at DigComp Foundation levels. DiGiUP, by contrast, was associated with higher demonstrated proficiency levels, particularly in information and data literacy, safety, and problem solving. The statistically significant differences between the two approaches, supported by large effect sizes, suggest systematic within-sample shifts in demonstrated competence. However, given the absence of a control condition and the exploratory design, these results should be interpreted as indicative of potential effectiveness rather than as definitive evidence of causal impact.

Addressing **RQ2** *(What is the added value of integrating game-based assessment and game-based learning?)*, the study provides exploratory evidence that combining GBA and GBL within a single, coherent pathway may offer practical benefits over treating assessment and learning as separate processes. Embedding assessment within gameplay appeared to reduce anxiety and encourage exploration and reflection among participants. In addition, trainer observations coded under A2 (Reduced fear of errors) and R2 (Shared emotional experience) suggest that participants perceived the games as non-threatening and collaborative. These findings illustrate how assessment can be experienced as part of a learning process in this particular implementation, though broader applicability requires replication in other contexts.

With respect to *RQ3 (How do motivational factors influence engagement with game-based digital skills training?)*, analysis of the Game-Play Questionnaire grounded in Self-Determination Theory indicates that the game-based activities were positively experienced in terms of autonomy, competence, and relatedness. High scores across all three dimensions, combined with acceptable to good internal reliability, suggest that motivational design played a meaningful role in participation engagement. These results reinforce theoretical claims that satisfaction of basic psychological needs is central to sustained engagement and learning.

Beyond addressing the research questions, the study contributes methodologically by demonstrating how DigComp 2.2 proficiency descriptors can be operationalised directly within game mechanics. This design-informed empirical work illustrates one way in which high-level competence frameworks can be translated into structured, learning interactions for older adults, offering a replicable approach for embedding competence-based assessment into accessible, low-pressure game formats. To complement this, all three SDT dimensions – Autonomy (A1-A3), Competence (C1-C3), and Relatedness (R1-R3) – were strongly reflected in the trainers' notes. Autonomy was evident in participants' willingness to make choices and take initiative; competence emerged through observable confidence gains; and relatedness was supported through peer interaction and shared emotional experiences.

Based on these pilot findings, several practice-oriented considerations may be relevant for researchers, educators, and designers working in similar adult-learning environments:

1. **Consider blended GBA-GBL pathways**: Rather than separating assessment from instruction, embedding competence evaluation within structured game activities may offer advantages in comparable facilitated settings.
2. **Align game design with competence frameworks**: Using established competence descriptors can support transparency and comparability, although effectiveness of such alignment should be further examined in other contexts.
3. **Attend to motivational design principles**: Treating motivational considerations (autonomy, competence, and relatedness) as integral to design, rather than peripheral features, may be particularly important when working with older learners.
4. **Use familiar and facilitated formats**: Board-game-inspired mechanics, small-group settings, and trainer moderation are well-received in these settings and appear to lower participation barriers.
5. **Combine quantitative and qualitative insights**: The integration of performance data with observational and experiential feedback provides a richer understanding of how learning unfolded during gameplay in this pilot study. Mixed-method approaches may therefore be valuable in similar exploratory evaluations.

While the pilot reported here was conducted in Slovenia, the intervention was developed and tested as part of the broader Erasmus+ DIGIBLEND project, which involved participants across six European countries: Slovenia, Serbia, Turkey, Italy, Germany, and Ireland. The consistency of the design principles: competence framework alignment, SDT-informed mechanics, and analogue-hybrid formats, across these diverse national and institutional contexts suggests that the approach is not limited

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to the Slovenian setting. Although a cross-country comparative analysis is beyond the scope of this paper, the multi-country implementation provides preliminary evidence of the approach's adaptability to varied cultural, linguistic, and institutional environments. Researchers and practitioners working in comparable adult education settings are encouraged to draw on the openly available game materials and evaluation instruments as a starting point for contextually adapted implementations.

6.2 Limitations and future work

While this study provides valuable insights into the use of game-based learning and assessment for digital competence development among older adults, several limitations should be acknowledged. First, the study was conducted with a moderate sample size (N = 45) drawn from three day-care centres in Slovenia, using a convenience sampling approach. Although this setting supports ecological validity, it limits the generalisability of the findings to other cultural, institutional, or national contexts. Future studies should involve larger and more diverse samples, including participants from different countries, socio-economic backgrounds, and adult education settings, to examine the robustness and transferability of the proposed approaches (Rasi-Heikkinen & Doh, 2023; Xu et al., 2025).

Second, the evaluation focused on short-term outcomes observed during and immediately after gameplay sessions. While the results demonstrate clear differences between INFINITY and DiGiUP in eliciting digital competence levels, the study does not assess long-term retention, transfer of skills to real-world contexts, or sustained behavioural change. Longitudinal research is therefore needed to investigate whether the observed competence gains persist over time and translate into everyday digital practices (Buragohain et al., 2023; Ji et al., 2026). Moreover, as the study did not include a control or alternative-method comparison, improvements cannot be attributed exclusively to the game-based interventions. Therefore, additional work is needed to incorporate control conditions to disentangle gameplay effects from facilitation and exposure.

Third, although DigComp proficiency levels were systematically mapped to game tasks, the assessment relied on performance within the game context, which may not fully capture participants' ability to apply skills independently outside the structured and facilitated environment. Future work could combine game-based assessment with external performance tasks, real-world simulations, or AI-based approaches to triangulate competence development (Buragohain & Chaudhary, 2025).

Fourth, the motivational experience was measured using a self-developed Gameplay Questionnaire grounded in Self-Determination Theory. While reliability analyses indicated acceptable to good internal consistency, the instrument was designed for this specific context and population. Future research should further validate and refine the GPQ, including exploratory and confirmatory factor analyses with larger samples, and examine its applicability across different game-based learning contexts and age groups (Rogers, 2024).

Finally, the study compared two specific game-based designs with distinct roles within a blended pathway. While this comparison highlights the complementary functions of baseline-oriented assessment and learning-focused gameplay, it does not exhaust the range of possible game mechanics, narratives, or facilitation strategies. Future work could explore adaptive difficulty mechanisms, personalised feedback, intergenerational gameplay formats, and fully digital or hybrid delivery models, including immersive methods (Buragohain et al., 2025). Moreover, it should investigate the role of facilitator training and group dynamics in shaping learning outcomes.

These limitations do not reflect a lack of scientific rigor or research quality point; rather, they point toward a rich agenda for future research that extends beyond proof-of-concept evaluation toward scalable, longitudinal, and cross-cultural investigations of game-based approaches for inclusive digital competence development in later life.

7. Conclusion

This study explored how game-based learning (GBL) and game-based assessment (GBA) can be designed and combined to support the development and evaluation of digital competences among older adults within a specific adult-education context. Through the design and implementation of two complementary game-based approaches – INFINITY and DiGiUP – and their pilot evaluation in facilitated small-group sessions, the study provides empirical evidence regarding the feasibility and potential value of such blended approaches. The findings indicate that INFINITY functioned effectively as a baseline-oriented assessment and confidence-building tool in this sample, enabling older adults to engage with digital competence evaluation in an accessible and low-pressure manner. DiGiUP was associated with upward shifts in DigComp proficiency levels, particularly in information and data literacy, safety, and problem solving. Distributional analyses and non-parametric statistical tests revealed statistically significant differences between the two approaches within the study sample.

Beyond performance outcomes, the study highlights the relevance of motivational design. Results from the Gameplay Questionnaire, grounded in Self-Determination Theory, suggested that the game-based activities were positively experienced in terms of autonomy, competence, and relatedness. These findings support the view that digital inclusion efforts may benefit from addressing not only competence acquisition but also learners' emotional and social experiences. From a theoretical perspective, the study illustrates one way in which competence frameworks such as DigComp 2.2 can be operationalised within concrete game mechanics, embedding assessment at the task level rather than treating it as a separate, standalone procedure.

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Methodologically, it offers a design-informed example of how assessment and learning can be integrated into a coherent game-based pathway tailored to older adult learners.

This work provides important, although context-bound, evidence that game-based approaches, when grounded in competence frameworks and motivational theory, can play a meaningful role in reducing digital exclusion among older adults. While further research is needed to explore long-term effects, scalability, and cross-cultural applicability, the results presented here provide a strong foundation for the continued development of inclusive, engaging, and evidence-based digital skills interventions for older adults.

Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Consent to participate

Ethics approval

This study involved human participants solely for the purpose of evaluating the effectiveness of the proposed methods and was conducted within the framework of the Erasmus+ DIGIBLEND project (project number 2021-1-DE02-KA220-ADU-000035280; https://digiblend.eu/en/the-project/). The ethical aspects of the project were assessed and approved by the European Commission at each formal stage of its lifecycle: the initial application, the mid-term review, and the final review.

All participation was entirely voluntary, with participants informed of their right to withdraw from the sessions at any time without penalty. No sensitive personal data, as defined under Regulation (EU) 2016/679 (General Data Protection Regulation – GDPR), was collected, processed, or stored. The activities fell outside the categories requiring formal ethics committee approval under the Slovenian Personal Data Protection Act (ZVOP-2, Official Gazette RS No. 163/22) and in accordance with the European Code of Conduct for Research Integrity (ALLEA, 2017).

As the research did not involve vulnerable populations in a manner posing foreseeable physical, psychological, or legal risk, and did not process personal or special categories of data under Article 9 of the GDPR, no further institutional or national ethics board approval was required.

Informed consent

Written informed consent was obtained from all participants and, where applicable, from their legal guardians or caregivers within each of the three day-care centres in Ljubljana, Slovenia, where the participants resided. The activity was carried out during the first week of April 2025, prior to the commencement of any testing activities. To ensure full comprehension of participation conditions and rights, the written consent was subsequently reviewed and confirmed verbally with each participant.

All participants were provided with clear and accessible information regarding: (i) the purpose of the study; (ii) their specific role and rights, including the right to withdraw at any time without consequence; and (iii) the intended use of their data. Data collection was limited to information required for statistical analysis and was processed exclusively in an anonymized form, in full compliance with Regulation (EU) 2016/679 (General Data Protection Regulation – GDPR).

A copy of the consent form template is included in the Supplementary Information accompanying this manuscript, in part A1.

Data availability

The data that support the findings of this study are made openly available and, due to reasons of sensitivity and privacy, have been anonymized for presentation of the statistical analysis and validation.

All data, code, statistical analysis, and materials based on which the results in this paper are obtained, are openly available (in anonymized form) in the following (anonymized for the double-blind peer review) GitHub repository: https://github.com/anonResearcher107/Game-based-learning

These include: the datasets containing the GPQ questions, the mapping of the game's questions and challenges to DigComp proficiency levels, the proficiency profile for each of the 45 participant, the pre-post testing results for both games and all statistical analysis based on it, including the code and the statistical analysis, as well as the data for the qualitative analysis based on the trainer notes, together with the coding of that data. In addition, the technical specifications, rules and the gameplay dynamics are also part of the study's GitHub. The content and the procedure for using all files is clarified in detail in the README.md file that is part of the repository.

Physically, data are located in controlled access data storage at the Laboratory for Open Systems and Networks at Jozef Stefan Institute, Slovenia, and are also available at a reasonable request from the corresponding author.

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Contribution of authors, as per the CRediT Taxonomy

	A.J.B	M.M	T.P
Conceptualization	X	X	X
Data curation		X	X
Formal analysis			X
Funding acquisition		X	
Investigation	X	X	X
Methodology		X	X
Project administration		X	
Resources	X	X	X
Software			X
Supervision		X	X
Validation	X		X
Visualization			X
Writing – original draft	X		X
Writing – review & editing	X	X	X

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